

— FINAL REPORT • EMBODIED AI & ADAPTIVE CONTROL

Torque, not position.

A student case study of **SATA** — Safe and Adaptive Torque-Based Locomotion Policies Inspired by Animal Learning. We *reproduce and analyse* it in Isaac Gym through the questions adaptive & robust control asks, then *carry the reproduction across engines* to Isaac Lab.

UNITREE G02

STAGE 1 · ISAAC GYM · PPO

STAGE 2 · ISAAC LAB · PHYSX 5

8 SEEDS EACH ENGINE

SUBJECT PAPER

Li et al., RSS 2025
marmotlab · NUS

STAGE 1 – ANALYSE

Reproduce → Ablate →
Stress-test → Compensate

STAGE 2 - PORT

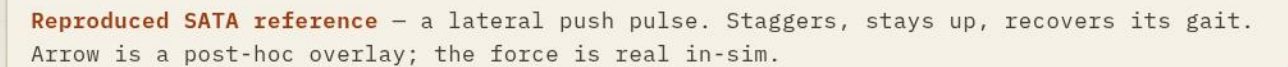
Carry the reproduction
Isaac Gym → Isaac Lab

PRESENTED BY

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[ name / course / date ]
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Torque control buys **compliance** — but raw torque is dangerous to hardware. SATA's bet: borrow the body's own safety limits from **biology**.

sim-to-real safety



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STAGE 01

Activation

First-order low-pass on torque sign — models muscle-recruitment delay, kills bang-bang switching.

STAGE 02

Hill model

Force-velocity law: max torque *drops* when the joint moves fast in the same direction.

STAGE 03

Fatigue

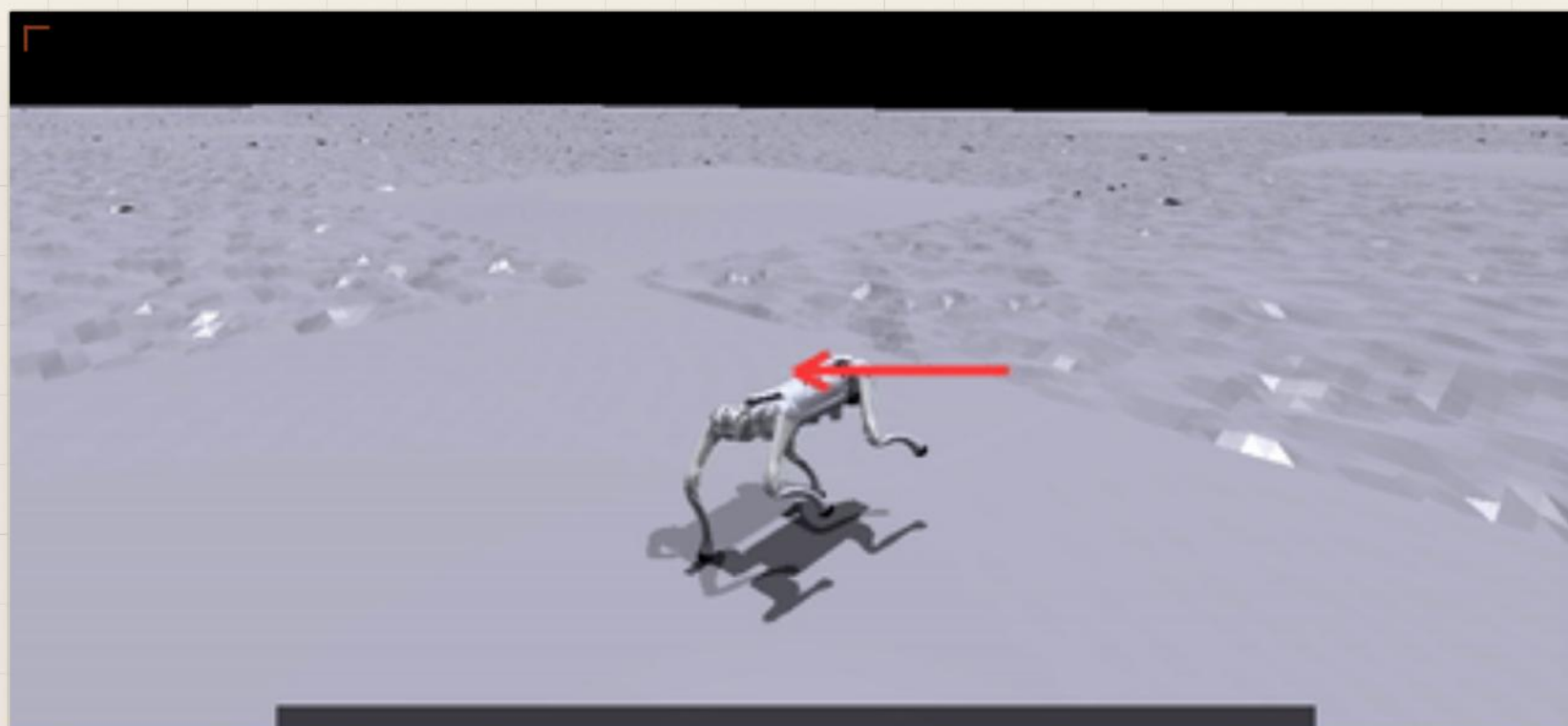
Leaky accumulator per joint, fed back into the observation. Small -0.05 reward penalty.

CURRICULUM

Growth

Gompertz ramp: front-leg torque ceiling 0.3→1.0, control rate 100→200 Hz.

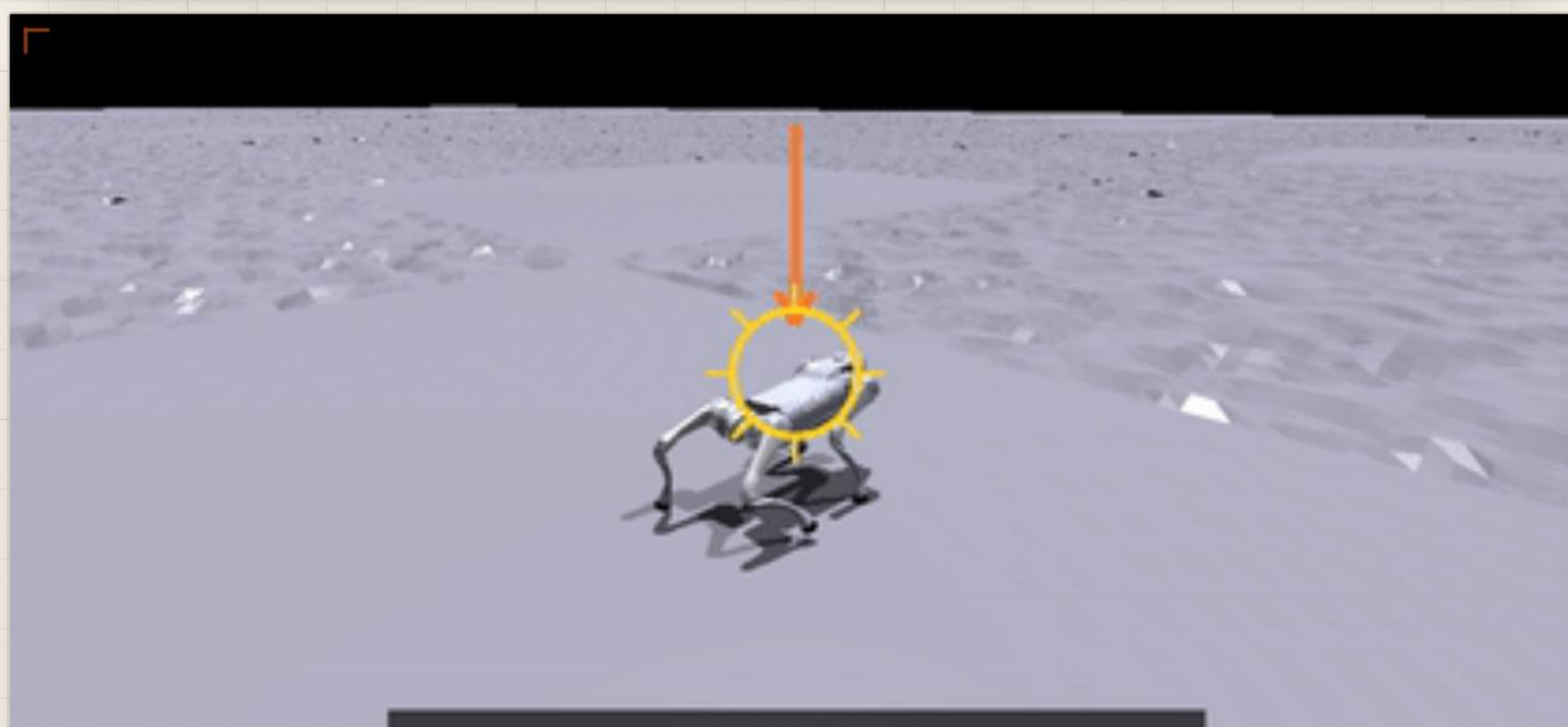
Source: [go2_torque.py](#) L221-248 (pipeline) · [go2_torque_config.py](#) L102-108 (toggles) · pinned commit 8fc422af



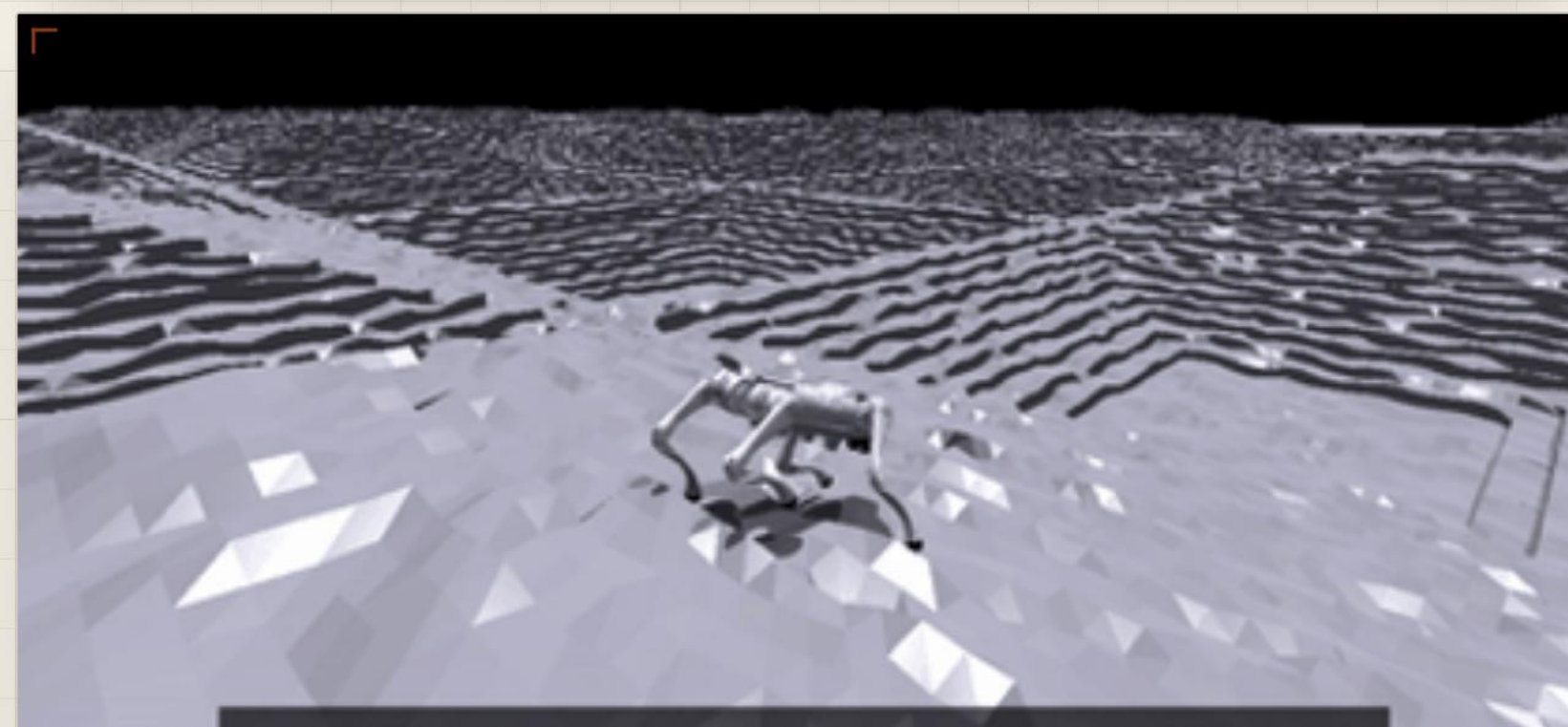
Lateral push - staggers, stays up, recovers.



Leg pulled (recoloured red) – resists an external tug.



Vertical impact – absorbed compliantly, then recovers.



Stairs - stair-trained hard_terrain policy.

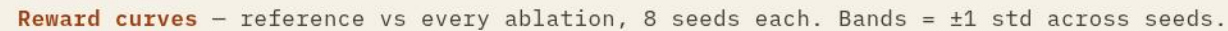
Disturbances are **real sim changes** (forces / friction / leg recolour); arrows · rings · labels are **post-hoc overlays** on the rendered frame, so an external force reads as external – not the robot thrashing.

104_{±16}

8

3000

The first 3 seeds gave 114 ± 6 . Extending to 8 revealed **genuinely larger seed variance** — including one late-training PPO collapse. We report the harder number.



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42.5 N·m

45 N·m

2.5x / 35x

On **the axis we measured**, the fatigue & activation constraints cost training reward but keep the policy **hardware-realisable** — they read here more like sim-to-real feasibility constraints than reward devices. That is *our interpretation on this scalar axis*, only one of the **two faces** the paper describes (it leads with exploration / trainability, which we do not measure here). Caveat: simulation only, no thermal model.



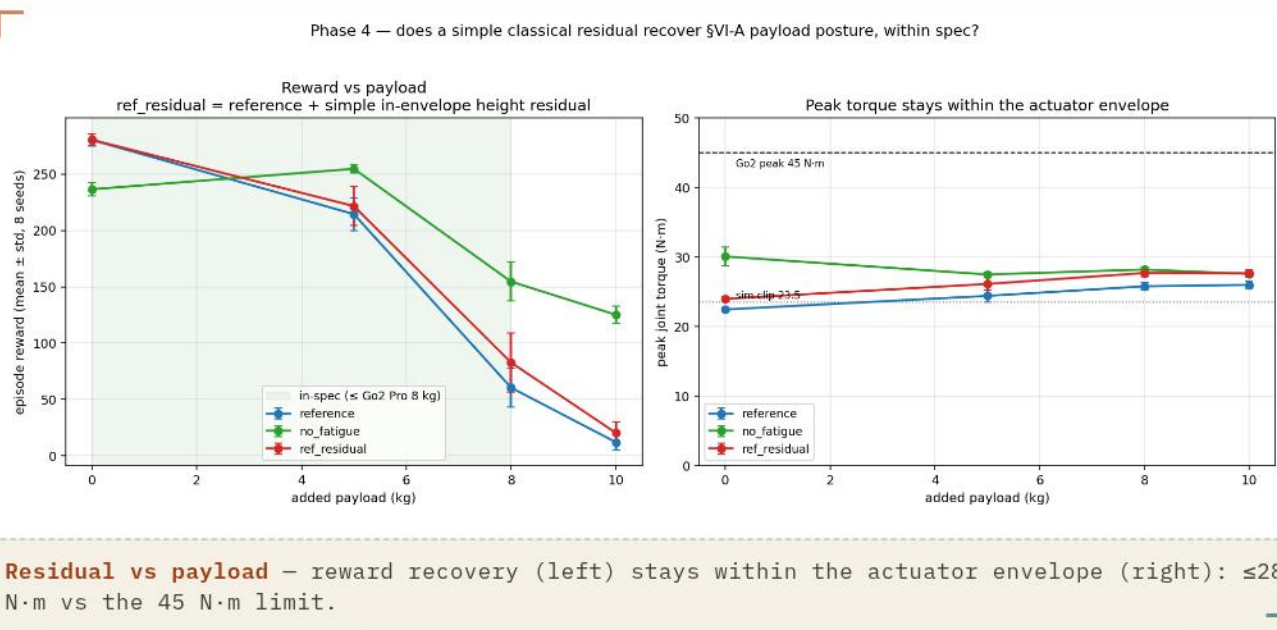
Both clips are **true to the data**: the reference reproduces the paper's own §VI-A payload limitation. "Staying up" is not automatically the win — staying up *within the envelope* is.

Can a classical bolt-on recover what safety trades away?

A stance-gated height-PD residual $\tau_{total} = \tau_{SATA} + \tau_{comp}$, added onto the **frozen** policy — every bio constraint intact, staying inside the 45 N·m envelope.

8 KG PAYLOAD	REWARD	Δ VS REF	P
reference	60.5	—	
+ residual	82.7	+37%	0.07 n.s.
no_fatigue	154.6	(ablation)	

A small **positive trend** (two-sided Welch $p \approx 0.07$, not significant) — and only $\sim \frac{1}{4}$ of the way to the ablation. The frozen policy *treats the residual as a disturbance*; that ceiling motivates co-trained RL + adaptation (RL2AC).



ADAPTATION QUESTION	CLASSICAL ANSWER	SATA'S ANSWER
Unknown dynamics	online parameter estimation	robust policy learned in sim
Disturbances	explicit compensation term	internal fatigue-state feedback
Generalization	stability proofs / guarantees	empirical, across terrains
Control output	analytic control law	neural torque policy

One caveat on that reading: the feasibility lens is what our scalar-reward axis can see. The paper itself *leads* with the bio mechanisms' role in exploration / trainability and only then their feasibility / smoothness role — so "feasibility constraints, not reward devices" is our axis-limited interpretation, not the paper's stated sole design intent.

τ **SATA** · torque-based adaptive locomotion · Gym → Isaac Lab

Stage 1 reproduced and analysed SATA in [Isaac Gym](#). As a second, smaller exercise we carry the **same reproduction** to [Isaac Lab](#) (PhysX 5, manager-based RL) — the robot held fixed (Go2) so the engine is the only thing that changes. The engine gap (contact, friction, actuator dynamics) doubles as a free **sim-to-real / robustness proxy** — a control-relevant question: does the adaptive policy hold when the dynamics shift underneath it?

PhysX 5 manager-based RL effort / torque actuators robot fixed: Go2



- rough — not flat — terrain
- reward = a single substep (not summed)
- variable-frequency residual carried over
- per-substep counters (episode + growth)
- calf velocity limit 15.70 → 20.07 rad/s
- SATA joint position limits written to the physics sim

ISAAC LAB · 8 SEEDS

ISAAC GYM REF · $\approx 104 \pm 16$ (TFEVENTS)



τ **SATA** · torque-based adaptive locomotion · Gym → Isaac Lab

Per-step decomposition removes the episode-length confound. The **task / tracking** terms match the Gym reference to $\sim 0.002/\text{step}$ — the port reproduces SATA's per-step task reward.

Limits (kept in view): we do *not* claim the engines are equivalent — this check is preliminary. The native Isaac Sim RTX render is blocked by a known driver-595 bug, so **every Isaac-Lab clip is a kinematic replay** of the Isaac-Lab-trained trajectory, rendered in Isaac Gym.

We have **not** separated "the penalty measures acceleration differently" from "the Lab gait is genuinely jerkier" — the matched task terms make the latter implausible, but we leave it an open question, not a verdict.

- **Stage 1 — reproduced & analysed.** The paper motivates the bio layers on **two faces** — it leads with early-stage exploration / trainability, then sim-to-real feasibility. Our scalar training-reward axis only sees the second: on that axis the bio layers read more like feasibility constraints than reward devices — *our axis-limited interpretation*, not the paper's sole stated intent. The data: removing bio raises clean-sim reward, yet the ablations leave the 45 N·m envelope.
- **Control-theory lens.** The Hill / activation ↔ feasibility analogy held on our data; "fatigue ↔ compensation" and "growth ↔ gain scheduling" did not.
- **A classical residual** recovers some lost payload capability within the envelope — modestly ($p \approx 0.07$, not significant).
- **Stage 2 — carried across engines.** A faithful Isaac Lab port (six fixes) reproduces SATA's per-step *task* reward; the reward-number gap is concentrated in one penalty term whose cross-engine definition differs — whether that fully explains it, vs. a jerkier Lab gait, is left open. This cross-engine check is **preliminary** — we do not claim the engines are equivalent.
- **Reproducibility carried over — including the failure mode:** a single late-training PPO collapse appears on *both* engines.
- **Limits in view:** simulation only · no thermal model · residual gains tuned on one seed · Isaac-Lab clips are kinematic replays (RTX render blocked) · the joint_acc gap is not yet isolated to measurement vs gait.

Cross-engine reproducibility

Reproduction & analysis of Li et al., *SATA* (RSS 2025) · Stage 1: github.com/Stanley-1013/bio-inspired-adaptive-locomotion · Stage 2: github.com/Stanley-1013/isaacsim-torque-locomotion

— REFERENCES & PROVENANCE

SUBJECT & CLOSEST PRIOR WORK

- | | |
|-----|--|
| [1] | SATA: Safe and Adaptive Torque-Based Locomotion Policies Inspired by Animal Learning. Li et al., <i>RSS 2025</i> . arXiv:2502.12674 · code: github.com/marmotlab/SATA @8fc422af |
| [2] | RL2AC: RL-based Rapid Online Adaptive Control for Legged Robot Robust Locomotion. <i>RSS 2024</i> . roboticsproceedings.org/rss20/p060 |
| [6] | Chen et al. Learning Torque Control for Quadrupedal Locomotion. <i>2022</i> . arXiv:2203.05194 |
| [7] | DecAP: Decaying Action Priors for Torque-Based Legged Locomotion. <i>IROS 2024</i> . arXiv:2310.05714 |

BIOMECHANICAL FOUNDATIONS

- [8] Hill, A. V. The heat of shortening and the dynamic constants of muscle. *Proc. R. Soc. B*, 1938. doi:10.1098/rspb.1938.0050
- [9] Liu, Brown & Yue. A Dynamical Model of Muscle Activation, Fatigue, and Recovery. *Biophysical Journal*, 2002. PMC1302927

LEARNING-BASED LOCOMOTION LINEAGE

- [3] Hwangbo et al. Learning agile and dynamic motor skills for legged robots. *Science Robotics* 2019. [arXiv:1901.08652](#)
- [4] Lee et al. Learning quadrupedal locomotion over challenging terrain. *Science Robotics* 2020. [arXiv:2010.11251](#)
- [5] Miki et al. Learning robust perceptive locomotion for quadrupedal robots in the wild. *Science Robotics* 2022. [arXiv:2201.08117](#)
- [10] Bellegarda & Ijspeert. CPG-RL: Learning Central Pattern Generators for Quadruped Locomotion. *RA-L* 2022. [arXiv:2211.00458](#)

REPRODUCTION STACK & PROVENANCE

- **Stage 1 • Simulator:** Isaac Gym Preview 4 • **RL:** PPO via `rs1_rl` • **Env:** `legged_gym`, Unitree Go2 `go2_torque` • 8 seeds × 6 conditions • 384 eval cells • Welch's t-test
- **Stage 2 • Engine:** Isaac Lab (PhysX 5, manager-based RL). isaac-sim.github.io/IsaacLab • Go2 torque/effort port • 8 seeds • github.com/Stanley-1013/isaacclab-torque-locomotion
- **Compute:** 4× RTX A6000 (lab K8s pod), both stages.
- **Stage 1 repo:** reproduction, analysis, plots & clips. github.com/Stanley-1013/bio-inspired-adaptive-locomotion
- **Companion docs:** [docs/sata-overview.md](#) • [control-perspective.md](#) • [REPRODUCTION_NOTES.md](#)

Stage-1 claims trace to `results/phase{1-4}/` (clips in `results/gym_reference/`) · Stage-2 numbers to `docs/REPRODUCTION_NOTES.md` · Isaac-Lab clips in `results/deck/` are kinematic replays.